

Advanced (Carolina Reaper)

- Q1.** Solve the system: $x^2 + y^2 = 25$, $y = x^2 - 4$. What is the solution set?
(A) (3, 5), (-3, 5)
(B) (3, 5), (-3, -5)
(C) (-1, -2), (1, 2)
(D) (-3, -4), (3, 4)
(E) (1, 1), (-1, -1)
- Q2.** A skateboarder rides $2x^2 + 5x - 3$ miles in x hours. If she rides for 2 hours, how many miles does she ride?
(A) 30 miles
(B) 15 miles
(C) 20 miles
(D) 25 miles
(E) 10 miles
- Q3.** Solve the system: $y = x^2 - 2x + 1$, $y = -x^2 + 4x - 3$. What is the solution set?
(A) (1, 0), (3, 3)
(B) (-1, 2), (1, 0)
(C) (2, 1), (-2, -1)
(D) (3, 1), (-1, -1)
(E) (1, 1), (-1, -1)
- Q4.** If $x^2 + y^2 = 16$ and $y = 2x - 3$, what is the value of x ?
(A) 1, -1
(B) 2, -2
(C) 1, 3
(D) -1, 3
(E) 2, 4
- Q5.** For the system $y = x^2 - 4$, $x^2 + y^2 = 25$, which point is a solution?
(A) (3, 5)
(B) (-3, 5)
(C) (-1, 3)
(D) (1, -3)
(E) (0, -4)
- Q6.** A store sells $x^2 + 3x - 2$ shirts per day. If the store is open for 5 days, how many shirts are sold?
(A) 50 shirts
(B) 75 shirts
(C) 80 shirts
(D) 60 shirts
(E) 100 shirts
- Q7.** Solve the system: $x^2 + y^2 = 16$, $y = 2x + 1$. What is the solution set?
(A) (-3, -5), (3, 5)
(B) (1, 3), (-1, -1)
(C) (3, 7), (-1, 1)
(D) (-1, 1), (1, 3)
(E) (2, 5), (-2, -1)
- Q8.** For the system $x^2 + y^2 = 25$, $y = x^2 - 2x + 1$, what are the possible values of y ?
(A) -2, 2
(B) 0, 4
(C) -1, 3
(D) 1, 5
(E) -3, 1

Open Ended Questions (FRQ)

- Q9.** The height h of a projectile in feet is given by $h = -16t^2 + 48t + 10$ where t is time in seconds. Find the time when the height is 30 feet.
- Q10.** Solve the system: $x^2 + y^2 = 20$, $y = x^2 - 4x - 5$. Find the solution set.

Chef's Tips

- Provide clear instructions to students
- Ensure students have necessary materials
- Monitor student progress closely